

**ADEKUNLE AJASIN UNIVERSITY,
AKUNGBA-AKOKO**

DEPARTMENT OF CYBERSECURITY

COS 202 ASSIGNMENT REPORT

**DEVELOPMENT OF A MATHEMATICAL
CALCULATOR (MC) AND PERSONAL POCKET
CGPA CALCULATOR (PPC) USING PYTHON**

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Submission Date: 5 July 2026

Introduction

This report presents the development of two Python applications: a GUI-based Mathematical Calculator and a Personal Pocket CGPA Calculator. The assignment demonstrates the use of Python, Tkinter, arithmetic operations, and selection control statements to solve practical computing problems.

Aim and Objectives

The aim of this assignment is to develop a Mathematical Calculator and a Personal Pocket CGPA Calculator using Python. The objectives are to design an interactive calculator, perform basic arithmetic operations, apply selection control statements, and accurately calculate CGPA.

Project One: Mathematical Calculator

The Mathematical Calculator was developed using the Tkinter library. It provides a graphical user interface with buttons for addition (+), subtraction (-), multiplication (*), division (/), modulus (%), exponentiation (^), Clear (C), and OFF. The program validates expressions and displays results interactively.

Project Two: Personal Pocket CGPA Calculator

The CGPA Calculator is a console-based application. It accepts the number of courses, course units, and letter grades. Using if, elif, and else statements, it converts grades to grade points, computes the CGPA, and displays the corresponding class of degree.

Testing and Results

Both programs were tested successfully. The screenshots are included separately in the repository as:

- Figure 1 – Screenshot (33).png: Calculator interface.
- Figure 2 – Screenshot (34).png: Calculator showing user input.
- Figure 3 – Screenshot (35).png: Calculator showing calculation result.
- Figure 4 – Screenshot (36).png: Personal Pocket CGPA Calculator showing both input and output.

Challenges Encountered

The major challenges encountered included designing a user-friendly interface, handling invalid input, and verifying the accuracy of arithmetic and CGPA calculations. These issues were resolved through testing and debugging.

Conclusion

The objectives of the assignment were achieved successfully. Both applications function correctly and demonstrate practical application of Python programming concepts.

References

1. Python Software Foundation. Python Documentation.
2. Tkinter Documentation.
3. COS 202 Lecture Notes.